HARP: Hop-Aware Selective Release for Privacy-Audited GNN Prediction APIs

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***Abstract*—Graph neural networks are increasingly served as prediction APIs: clients query a node and receive a softmax score vector used for ranking, routing, and analytics on graphs from thousands to millions of nodes. At this scale the API is a continuous data product—operators must keep accuracy high, keep most responses bit-exact for caching and compliance, and remain *audit-compliant* under membership inference checks. Uniform Laplace posterior perturbation (LBP) cannot meet that joint goal: any $b>0$ perturbs *every* response, so a clean-majority constraint is infeasible by construction. We formalize this as *constrained score release*—maximize accuracy subject to a membership audit and a minimum exact-response fraction—and propose HARP, a selective release framework for GNN score APIs. HARP is not proposed as a stronger membership defense than uniform noise; it is the mechanism that can satisfy a clean-majority constraint while remaining audit-compliant and recovering accuracy versus published strong LBP. HARP ranks nodes (topology or audit-guided vulnerability), expands high-risk seeds by k hops, and applies a protector only on that set, with Frac/Mass dials and a session budget against re-query averaging. Across six graphs, GAT/inductive checks, ogbn-arxiv ($\sim$169k) with defense-aware LiRA, and an ogbn-products subsampled audit ($\sim$15k of 2.4M), HARP recovers large accuracy drops versus strong LBP while enabling a 60% clean majority that uniform noise cannot provide.**

***Index Terms*—graph neural networks, prediction APIs, Big Data systems, selective release, calibration, membership inference**

I. Introduction

A. Motivation

Graph neural networks (GNNs) are widely deployed as *prediction APIs*: a client queries a node ID and receives a softmax score vector [9], [10]. Those scores drive ranking, triage, and analytics. They also leak whether a node was in the training set, via membership inference attacks (MIAs) [1], [2], [4], [5].

At Big Data scale the API is not a one-shot model export. It is a stream of released probabilities. Operators therefore face three systems concerns: **volume** (graphs from 10^3 to 10^6+ nodes), **velocity** (clients may re-query randomized answers), and **veracity** (many pipelines need stable, well-calibrated scores for caching, A/B tests, and compliance).

Research conducted with the Data Science Laboratory (DiSC), University of Nevada, Las Vegas.

The usual graph defense is Laplace posterior perturbation (LBP) [5]. Strong uniform noise (b=0.3) lowers attack success but often collapses accuracy. Weaker noise keeps accuracy but still perturbs *every* node and raises expected calibration error (ECE). Neither option yields a *clean majority* of exact scores—a concrete data-quality property for downstream consumers.

The systems question is therefore **constrained budget allocation**: maximize task accuracy subject to a membership audit *and* a minimum fraction of bit-exact responses.

a) Main idea.: Membership signal on graphs is not node-local: message passing mixes neighborhoods. HARP (Hop-Aware selective Release Policy) ranks nodes with a cheap topology estimator (LTE) or an audit-guided vulnerability score, expands high-risk seeds by k hops into a protected set $\mathcal{P}_k$, and applies a protector (strong Laplace or selective masking) *only* on that set. Under our locked setting (protected fraction Frac=0.40), about 60% of API responses stay bit-exact—impossible for any uniform $b>0$.

B. Gap

Prior defenses include LBP [5], MemGuard [6], train-time GNN MIA methods (GTD, MaskArmor) [21], [22], AdvReg [15], and training-time DP-GNNs [13], [14], [27], [36]. MemGuard and uniform LBP both perturb every response; train-time methods do not expose Mass/Frac release accounting. None jointly provide a hop-expanded release set, constructor×protector policy grid, Mass/Frac dials, matched-budget calibration, and release complexity under standard membership audits. HARP targets that systems gap.

a) Running example.: On Cora (n=2708), strong LBP cuts accuracy from ≈0.88 to ≈0.71. Weak uniform noise preserves accuracy but perturbs every score (ECE ≈0.16). HARP matches that weak-noise budget, keeps ≈60% of scores exact, reaches ECE ≈0.033 and Acc ≈0.83, and only modestly changes membership-attack AUROC. The operator gains usable accuracy and reproducible scores for most queries—not a claim that clean nodes are "membership-safe."

b) Non-goal (stated up front).: HARP is *not* a claim of lower population LiRA than equal-mass or strong uniform LBP. When ExactFrac= 0 is allowed, uniform noise is often the better Acc–LiRA point. HARP's claim is narrower and systems-shaped: it is the feasible class of policies under ExactFrac$\geq c>0$, with hop-consistent allocation, operator dials, and Acc/calibration recovery versus the published strong-LBP recipe.

C. Contributions and paper map

1) **Constrained score release.** Formalize maximize Acc s.t. LiRA$\leq \tau$ and ExactFrac$\geq c$; show uniform LBP is infeasible for $c>0$ (Remark 1); prove hop-consistency is necessary under one-hop score mixing (Prop. 1) and give selective local-ε accounting (Prop. 3).
2) **Selective release framework.** Constructor$\times$protector$\times$Mass/Frac$\times$session budget B, with $O(|E|+n\log n)$ setup (Prop. 2).
3) **Systems headline.** Versus strong LBP: $+12.5$ Acc on Cora at Frac$=0.40$; versus equal-mass: a 60% clean majority and $\sim 5\times$ lower ECE *on Cora*; quantified cache-hit simulation for ExactFrac; portable clean-slice fidelity.
4) **Scale audits.** ogbn-arxiv Acc recovery with defense-aware LiRA ($n_{\mathrm{sh}}\in\{2,4\}$); ogbn-products systems probe plus subsampled LiRA on a 15k-node induced subgraph [28].

Section II related work; Section III threat model; Section IV HARP; Sections VI–VII evaluation; Section VIII discussion; Section IX conclusion.

II. Background and Related Work

A. Membership inference on classical models

Shadow-model membership inference [1] and simple confidence features [2] show that prediction APIs leak training membership [3], [33]. Related extraction and inversion attacks [25], [26] further motivate treating scores as a privacy surface. Likelihood-ratio attacks (LiRA) are the current audit standard [4]. We use LiRA as an *operator audit* of whether release noise is warranted—not as the sole paper objective. We also track calibration because overconfident scores are a recurring membership cue [17], [34].

B. Release-time defenses on scores

MemGuard [6] adversarially perturbs confidence vectors to confuse attack classifiers while preserving the top-1 label; prediction purification similarly sanitizes released scores. Both act on *every* response and therefore cannot satisfy a clean-majority constraint. LBP [5] is the graph-native uniform Laplace baseline. HARP differs by selecting a hop-consistent minority and leaving the rest bit-exact.

C. Membership inference on graphs

Olatunji et al. [5] demonstrated node-level black-box MIAs on GNNs and proposed LBP and neighborhood sampling defenses. He et al. [38] systematically study node-level MIAs under varying topology knowledge. Subsequent work studies graph-level membership [7], embedding leakage [24], and broader attack taxonomies [8]. GTD [21] targets the transductive setting with alternate training and loss flattening. Khosla [39] shows that training-graph construction and inference-time edge access directly modulate membership advantage—orthogonal to our score-API audit protocol. GPS [35] studies structure→*attribute* leakage; we target score-based *membership*. Our threat model matches the score-based black-box node setting; we do not claim white-box, structural-modification, or federated attack coverage.

D. Architectures, regularization, and privacy

GCNs [9] apply spectral neighborhood averaging; GraphSAGE [10] uses mean aggregation and is often more robust under heterophily. DropEdge [11] randomly deletes edges during training. DP-SGD and DP-GNNs [13], [14] provide formal privacy guarantees at training time. Homophily modulates the train/test ϕ-gap [18]–[20]: under low h and sparsity, supervised memorization widens membership cues—motivating risk-weighted budget allocation.

E. Positioning

Our contribution is a **selective release framework** for GNN score APIs with Mass/Frac systems metrics. Uniform output noise (LBP, MemGuard) and train-time defenses (GTD, MaskArmor) are baselines. Formal DP (GAP) remains the right tool when an ε guarantee is required [13]; we do not re-implement GAP under our splits. In short: LBP/MemGuard spend on every node; GTD/MaskArmor harden training without release Mass/Frac; GAP answers formal ε; HARP answers constrained exact-fraction release under audits.

a) Systems lens (why Big Data).: Training systems optimize memory and throughput. Score-API release is an adjacent surface: every prediction is a data product with a utility and quality cost. HARP logs Frac and Mass the way operators already log batch size. We compare against the published strong LBP recipe (b=0.3) and against equal-mass weak noise so reviewers see what the same budget buys when spent uniformly.

III. Problem Setup

A. Notation

Let $G=(V,E)$ be an undirected graph with $n=|V|$ nodes, feature matrix $X\in\mathbb{R}^{n\times d}$, and labels $y\in\{0,\dots,C-1\}^n$. A random split partitions V into supervised train $\mathcal{T}$, validation $\mathcal{V}$, and test $\mathcal{Q}$ with ratios $40\%/20\%/40\%$ (resampled per seed). Training is *transductive*: every node participates in message passing; only $\mathcal{T}$ enters the cross-entropy loss. We write $\hat{p}_v$ for the softmax posterior at v, r_v for a risk score in $[0,1]$, $\mathcal{P}_k$ for the hop-expanded protected set, Mass$=\sum_v \sigma_v$, and Frac$=|\mathcal{P}_k|/n$.

B. Constrained score release

Operators rarely optimize Acc alone. They need usable accuracy *and* a reproducible majority of API responses, subject to a membership audit.

Definition 1 (Constrained Score Release)**.** *Given a trained GNN and release channel, choose a release policy* Π *to*

$$\max_{\Pi} \ \mathrm{Acc}(\Pi) \quad \textit{s.t.} \quad \mathrm{LiRA}(\Pi) \le \tau, \quad \mathrm{ExactFrac}(\Pi) \ge c,$$

where ExactFrac is the fraction of nodes whose released posterior equals $\hat{p}_v$ *bit-exactly,* $\tau \in [0.5, 1]$ *is an audit threshold, and* $c \in [0, 1]$ *is the required clean fraction.*

Remark 1 (Uniform LBP is infeasible for $c>0$)**.** *Independent Laplace noise with scale* $b>0$ *on every coordinate changes the value almost surely, so ExactFrac*$=0$ *for any uniform LBP policy. Selective release with protected fraction Frac achieves ExactFrac*$= 1 - \mathrm{Frac}$ *by construction.*

This reframes the Acc–LiRA comparison. Equal-mass LBP can win Acc subject to LiRA$\le \tau$ when $c{=}0$ is allowed; HARP solves the strictly harder problem with $c{>}0$. The Frac sweep traces HARP's Pareto frontier under that constraint.

C. Threat model

The adversary is a black-box client: it queries softmax scores and tries to guess whether a node was in the training set [2], [4], [5]. It does not see gradients or embeddings. White-box, link-stealing, and label-only APIs are out of scope [8], [12].

We also stress three practical variants: defense-aware shadows, multi-query averaging ($K \in \{1, 5, 20\}$), and an LTE-aware query adversary that focuses on clean nodes, protected-set boundaries, and top-decile LTE seeds (Sec. VII-C0k). We do not claim robustness to poisoning or unbounded queries without a session budget.

D. Attack features (brief)

Following Salem-style score attacks [2], we use confidence, true-class probability, entropy, and a margin term as ϕ-features. These are the signals membership audits and optional train-time alignment target.

E. Metrics

Utility: test accuracy (primary), plus F1/AUROC where useful. **Membership audit:** we report LiRA AUROC [4] and TPR at $0.1\%/1\%$ FPR—how well an attacker separates train members from held-out nodes (0.5 = chance). We also report confidence-based attacks for continuity with prior GNN work [2], [5]. **Data quality:** ECE [17], clean-slice ECE, and the clean-score majority $(1 - \mathrm{Frac})$. LiRA tells operators *whether* to spend noise; Acc and ECE tell them *what they paid* in utility and score fidelity.

Remark 2. *The primary protocol is transductive (non-members remain in the graph during training), matching common semi-supervised GNN APIs. We additionally report an inductive edge-exclusion check on Cora (Sec. VII-C0l) where test-incident edges are dropped at train/infer time.*

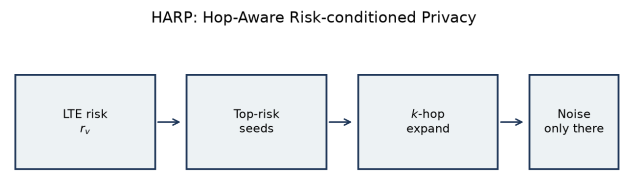

Fig. 1. HARP pipeline: risk ranking → high-risk seeds → k-hop expansion → protector only on the protected set.

IV. HARP: Hop-Aware Selective Release

A. Design principle

Uniform posterior noise wastes budget on low-risk nodes and cannot meet ExactFrac$\ge c{>}0$. Node-local risk weighting is still incomplete for GNNs: if v is protected but a neighbor u is released cleanly, aggregation can reintroduce membership signal into u's score. HARP therefore builds a hop-consistent protected set inside a small policy grid (Fig. 1):

1) **Constructor:** rank nodes with LTE (topology), audit-guided vulnerability, or random risk.
2) Pick high-risk seeds (or search the seed fraction to hit a target Frac).
3) Expand $\mathcal{P}_k = \mathrm{Ball}_k(\mathcal{R})$ on the undirected graph.
4) **Protector:** strong Laplace on $\mathcal{P}_k$, or selective masking (exact on clean; one-hot top-1 on $\mathcal{P}_k$).
5) Optionally train with risk-weighted ϕ-alignment on the same set; enforce session budget B at serving time.

Locked HARP uses LTE + Laplace at Frac$=0.40$; audit seeds and masking are framework ablations.

a) Noise mass and clean fraction.: $\mathrm{Mass} = \sum_v \sigma_v$ and $\mathrm{Frac} = |\mathcal{P}_k|/n$ are operator accounting metrics. Uniform LBP at scale b has Mass$=bn$ and Frac$=1$. See paragraph IV-D0a for the locked-budget identity.

b) Locked-config example (Cora).: On Cora ($n{=}2708$), Frac$=0.40$ protects $\approx$1083 nodes after one-hop expansion; $\approx$1625 nodes stay exact. $\mathrm{Mass}_{\mathrm{HARP}}\approx$325 matches equal-mass LBP with $b{=}0.12$. Strong LBP ($b{=}0.3$) spends Mass$\approx$812 on all nodes—the Acc-collapse baseline in Table I.

c) Release transform (Laplace protector).: Let $\sigma_v = \sigma_{\mathrm{strong}} \mathbb{1}[v \in \mathcal{P}_k]$. The released posterior is

$$\tilde{p}_v = \mathrm{renorm}\Big(\big[\hat{p}_v + \mathrm{Lap}(0, \sigma_v)\big]_+\Big), \qquad (1)$$

with independent Laplace coordinates and the same transform on every defense-aware shadow. The masking protector instead returns $\hat{p}_v$ for $v \notin \mathcal{P}_k$ and a one-hot of $\arg\max \hat{p}_v$ for $v \in \mathcal{P}_k$.

d) Algorithm (systems view).:

1) **Risk:** $r \leftarrow \mathrm{Constructor}(G, \mathcal{T})$ in $O(|E|)$ (LTE) or via a small shadow ensemble (audit).
2) **Seeds:** binary-search seed fraction so $|\mathrm{Ball}_k(\mathcal{R})|/n \approx \mathrm{Frac}^\star$.
3) **Expand:** $\mathcal{P}_k \leftarrow \mathrm{Ball}_k(\mathcal{R})$ via BFS ($O(|E|)$).
4) **Train (optional):** CE + risk-weighted AdvReg/MMD with $r_v \cdot \mathbb{1}[v \in \mathcal{P}_k]$.

5) **Release:** apply (1) (or the masking protector); log $(\mathrm{Mass}, \mathrm{Frac}, \mathrm{Acc}, \mathrm{LiRA}, \mathrm{ECE})$.

B. Risk constructors

LTE uses only G, $y|_{\mathcal{T}}$, and the train mask:

$$s_v^{\mathrm{deg}} = \frac{1}{1+\deg(v)}, \quad s_v^{\mathrm{het}} = 1 - \hat{h}_v, \quad s_v^{\mathrm{sup}} = \frac{|\mathcal{N}(v)\cap\mathcal{T}|}{\max(1,\deg(v))}, \tag{2}$$

with $\hat{h}_v$ the train-neighbor label agreement. Raw r_v is the mean of the three terms, min–max normalized. LTE is a cheap correlative proxy—not an oracle that predicts population LiRA alone.

Audit-guided ranking (HARP-audit) reuses the shadow ensemble already needed for LiRA: for each node, score $|\mu_{\mathrm{IN}} - \mu_{\mathrm{OUT}}|$ of logit-true-class confidence across shadows (Aerni-style vulnerability [37]), then seed and hop-expand at matched Frac. When shadows are affordable, audit ranking is the preferred constructor; LTE remains the no-shadow approximation.

Random ranking is the matched-Frac control isolating topology/audit signal from selectivity itself. Empirically, LTE beats random on LiRA for heterophilic Chameleon/Actor (Table IX); audit ranking targets the Cora case where LTE loses LiRA. Frac/Mass remain the primary population levers; the constructor is a ranking prior inside a fixed budget.

C. Why k-hop expansion

Let $\mathcal{R}$ be seeds. A GNN score at u already mixes information from $\mathrm{Ball}_k(u)$ before any release noise is applied. If a membership-sensitive seed $v \in \mathcal{R}$ is noised at release but a neighbor u is returned bit-exact, the adversary simply queries u.

Proposition 1 (Hop-consistency necessity under one-hop mixing)**.** *Fix a one-hop score model $s_u = \alpha\, z_v + (1-\alpha)\, z_u$ for $u \in \mathcal{N}(v)$, $\alpha \in (0,1]$, where z_v carries a membership cue of gap $\gamma > 0$ between member and non-member worlds, and release adds independent Laplace noise only to chosen nodes' scores. If the release policy noises v but leaves u exact, then any adversary that observes s_u retains advantage equal to the unprotected leaf advantage. Noising* $\mathrm{Ball}_1(\{v\})$ *is necessary and sufficient to place Laplace noise on every score that linearly depends on z_v under this mixer.*

Proof sketch. Under seed-only protection the released leaf score equals the clean mixer output, so the member/non-member gap in $\mathbb{E}[s_u]$ remains $\alpha\gamma$. Expanding protection to u adds $\mathrm{Lap}(0,\sigma)$ to s_u, which does not change the gap in expectation but is the unique one-hop cover that subjects every α-dependent coordinate to noise. A synthetic mixer confirms the ordering: unprotected and seed-only leaf AUROCs stay near 1.0, while hop-1 protection lowers the leaf attack to $\approx$0.72 (Fig. 5, left)—still above chance, but no longer a near-perfect membership channel. □

Frac—not k—is the population Acc–Mass knob; k is a coverage constraint on protected-set geometry. We keep k=1 as the default for 2-layer SAGE/GCN; $k\in\{0,2\}$ appear in the consolidated ablation (Table IX).

D. Complexity and accounting

Proposition 2 (HARP setup and release complexity)**.** *Let $G = (V,E)$ have $n = |V|$ nodes and $m = |E|$ adjacency endpoints, with C classes. Fix hop radius $k = O(1)$ and $T = O(1)$ binary-search iterations for a target protected fraction. Then setup (LTE + seeds + k-hop expand) runs in $O(m + n\log n)$ time and $O(n+m)$ space. Release runs in $O(nC)$—matching uniform LBP's tier, with less noise-sampling work whenever* $\mathrm{Frac} < 1$.

Proof sketch. LTE scans edges then normalizes n scores: $O(m{+}n)$. Seed selection is $O(n\log n)$ plus T BFS expansions at $O(m+n)$. Release writes scales, samples Laplace on $\mathcal{P}_k$, and renormalizes: $O(nC)$. □

Proposition 3 (Selective local privacy accounting)**.** *Interpret each node's clean posterior as a query with ℓ_1-sensitivity at most Δ=2 (worst-case change on the simplex). Independent Lap(σ) on every coordinate of $v \in \mathcal{P}_k$ yields $\varepsilon_{\mathrm{loc}} = \Delta/\sigma$ local DP for that node's release. Nodes in $V \setminus \mathcal{P}_k$ have ε=∞. Hence any policy with ExactFrac$>$ 0 has infinite* global *ε: ExactFrac$>$ 0 and global DP are incompatible by construction.*

At the locked point (σ=0.30, Frac=0.40) this gives $\varepsilon_{\mathrm{loc}}$=2/0.3 $\approx$ 6.67 on the protected minority and ε=∞ on the 60% clean majority. HARP therefore provides a finite local guarantee where it spends noise, and an explicit non-guarantee where it keeps scores exact. When a legal global ε is required, use training-time DP (e.g., GAP) instead [13].

a) Accounting identity (stated once).: Under locked Frac and fixed σ_{strong}, $\mathrm{Mass}_{\mathrm{HARP}} = \mathrm{Frac} \cdot n \cdot \sigma_{\mathrm{strong}}$ by construction—an accounting identity, not an empirical finding. The empirical contributions are (i) Acc recovery at fixed strong σ versus published LBP b=0.3, and (ii) calibration and ExactFrac at matched Mass versus weak uniform noise. We do not restate this caveat elsewhere.

E. What HARP is not

HARP is not differential privacy [27] and is not a claim of lower population LiRA than uniform noise under ExactFrac= 0. It is an empirical selective release framework for the constrained problem: spend a protector on a hop-consistent minority, keep the rest exact, and audit with membership attacks.

a) Intuition.: Think of HARP as a two-tier API. Low-risk nodes return the model's exact softmax. High-risk nodes (and their one-hop neighbors) get strong Laplace—or label-only masking—on a minority of the graph. The constructor ranks *where* to spend; Frac sets *how much*; the session budget limits re-query averaging.

V. Motivation: When Selective Privacy Pays

Matched-budget undefended LiRA (Table XII) concentrates leakage in small/mid heterophilic cells; several mid/large graphs sit near chance under n_{shadows}=4. That audit motivates selective spend: do not pay strong uniform noise where LiRA≈0.5; deploy HARP where LiRA≫ 0.5 *or* where veracity (exact/calibrated scores) is required under a strong-σ policy.

a) Three deployment regimes.: Heterophilic graphs (Chameleon/Actor) show auditable LiRA and LTE helps at fixed Frac; homophilic Cora/Citeseer emphasize veracity at matched Mass; large/audit-null graphs (PubMed/Photo/ogbn) often warrant Frac=0 unless policy forces strong σ (ogbn Acc recovery +0.420).

VI. Experimental Protocol

A. Pre-declared rules

Primary privacy = LiRA AUROC; primary utility = test Acc; systems = noise mass, protected fraction, and clean-score majority (1 − Frac). Hyperparameters are not selected to minimize *test* LiRA. All HARP comparisons against LBP use the published strong scale b=0.3 as the primary baseline; equal-mass b=0.120 is a mandatory fairness control, not the headline recipe.

B. Systems metrics (definitions)

We log four operator-facing quantities alongside Acc and LiRA:

- **Mass** = $\sum_v \sigma_v$: total Laplace scale spent at release (auditable noise budget).
- **Frac** = $|\mathcal{P}_k|/n$: fraction of nodes that receive strong noise.
- **Clean majority** = 1 − Frac: share of API responses returned bit-exact.
- **ECE**: whether released probabilities match empirical frequencies (lower is better calibrated).

Frac is the main dial: at fixed strong scale σ_{strong}, increasing Frac spends more Mass and perturbs more nodes, usually improving LiRA but hurting Acc and reducing the clean majority (Table VI). These metrics are cheap to log alongside Acc/LiRA and make privacy spend auditable the way train-seconds and peak RSS make training spend auditable.

C. Locked HARP

λ=0.5, k=1, σ_{strong}=0.30, target Frac=0.40, HCAG on for GraphSAGE/GCN, train-on-protected, warmup 5 (Cora val Acc; not test LiRA). Baselines use LBP b=0.3 / equal-mass b=0.120; GTD γ=1.0; MaskArmor top-k masking; MemGuard max-ℓ_1=0.2; DP-SGD Acc-tuned (vacuous ε); HARP $k \in \{0, 1, 2\}$, audit/random seeds, and selective masking at the same Frac. GAT runs use the same Frac/σ lock with HCAG off.

D. Baselines and attacks

`none`; strong LBP; equal-mass LBP; MemGuard [6]; GTD [21]; MaskArmor [22]; clip+noise DP-SGD; HARP and hop/Frac/constructor/protector ablations. Gaussian LiRA with n_{shadows}=4 (primary); we also sweep $n_{\text{shadows}} \in \{4, 16, 64\}$ on Cora/Chameleon for stability. Defense-aware shadows share the release transform. Five seeds with paired bootstrap CIs on Cora fairness/Frac; three seeds elsewhere unless noted. ogbn-arxiv uses NeighborLoader, n_{shadows}=2, gate off, warmup 3 (cost); Frac/σ remain locked.

E. What each baseline tests

We include defenses that isolate different design choices rather than maximize row count. **Strong LBP** (b=0.3) is the published recipe operators would copy; it anchors the Acc-collapse problem. **Equal-mass LBP** (b=0.12) matches HARP's Mass at Frac=0.40 and answers: "what if we spent the same budget uniformly on every node?" **MemGuard** is a simplified MemGuard-style baseline (random-search perturbation preserving argmax; not a full adversarial-training reproduction of Jia et al.): it perturbs every score, so ExactFrac= 0—the natural competitor when c=0 is allowed. **Selective masking** is the simpler protector alternative (exact on clean; label-only on $\mathcal{P}_k$). **GTD** and **MaskArmor** are train-time GNN MIA defenses; they harden training rather than add release-time Laplace, so Mass/Frac are not applicable (Table I, "—"). **Clip+noise DP-SGD** provides a formal-privacy anchor (with vacuous ε at our Acc-tuned point). **HARP ablations** (k, release-only, random/audit seeds, masking) attribute Acc and LiRA effects to hop expansion, training alignment, constructor, and protector respectively.

F. Attack protocol (summary)

We train defense-aware shadow models, release transformed posteriors, and report LiRA AUROC with ϕ-features [2], [4]. Validation nodes are excluded from membership labels; test Acc/ECE use the held-out test split. Multi-query experiments average $K \in \{1, 5, 20\}$ independent draws; session policies cap fresh draws at B=5.

G. Negative controls

PubMed (near-chance undefended LiRA): selective privacy should not be forced to spend a full uniform budget; HARP still preserves Acc vs. strong LBP at Frac=0.40 (Acc 0.839 vs. 0.768). ogbn-arxiv / ogbn-products: near-chance or memory-bounded audits under matched budgets—audit first, harden second [28].

VII. Results

A. Overview

We organize results around operator questions: **(Q1)** Can HARP keep accuracy and score quality at a matched noise budget? **(Q2)** Does it recover Acc versus strong uniform LBP? **(Q3)** How do Frac, multi-query, and session budgets behave? **(Q4)** Do patterns hold across architectures, inductive splits, and larger OGB graphs? Equal-mass LBP remains the honest Acc–membership baseline when exact scores are not required.

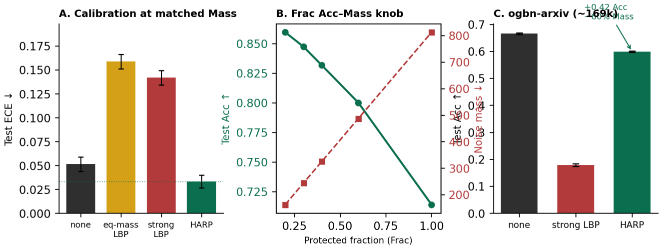

Fig. 2. Systems headline. (A) ECE at matched Mass. (B) Frac Acc–Mass dial. (C) ogbn-arxiv Acc vs. strong LBP.

B. How to read the trade space

Three columns answer different goals. **Acc** is task utility. **LiRA** is a membership audit (lower is harder to attack). **ECE / clean majority** is data quality for downstream consumers. Equal-mass LBP and MemGuard can look competitive on Acc–LiRA while still noising every response (ExactFrac$=0$). HARP's headline is Acc recovery plus exact/calibrated scores under ExactFrac$\geq c$ (Remark 1). Pick the constraint first—especially whether $c>0$ is required—then read Tables I–IV.

C. HARP systems headline: accuracy and score quality at matched Mass

a) Primary claim.: Fig. 2 and Table I: at matched Mass$\approx$325 on Cora, equal-mass weak LBP (b=0.120) perturbs every node (ECE 0.159). HARP yields ECE 0.033 with Frac=0.40—nearly 5$\times$ better calibration *on Cora*—and a 60% bit-exact clean majority. Equal-mass cannot produce exact scores; on some other graphs weak uniform noise can look competitive on population ECE while still noising every response (Table IV).

b) Utility vs. published strong LBP.: Against LBP b=0.3 [5], HARP raises Acc 0.707 $\to$ 0.832 (ΔAcc +0.125) at LiRA 0.517 $\to$ 0.533 and Frac=0.40 (Tables I, III). The operational win is accuracy at the same strong per-node noise on a minority of nodes. The pattern repeats on Citeseer/Chameleon/Actor/PubMed/Photo (Table III).

c) Membership audit (honest secondary).: At matched Mass, weak uniform LBP can beat HARP on Acc–LiRA (Cora Acc 0.870 / LiRA 0.520). If the only goal is Acc subject to LiRA$\leq\tau$ and exact scores do not matter, prefer equal-mass LBP. HARP is for operators who also need ExactFrac$\geq c>0$—not a stronger membership defense than uniform noise.

GTD and MaskArmor rows report Acc/LiRA/ECE only: both defenses modify training, not release posteriors, so they have no Mass or Frac to compare against HARP's release budget. MemGuard preserves Acc but leaves ExactFrac$=0$ and only modestly reduces LiRA under our audit. Selective masking keeps ExactFrac$=0.60$ and Acc high, but *raises* LiRA (clean scores still leak; one-hot protected scores remain confident)—justifying the Laplace protector when the audit requires it.

Slice ECE localizes the win on Cora (5 seeds): $\text{ECE}_{\text{clean}}$=0.018 vs. ECE_{prot}=0.081 (population 0.033). Clean nodes match undefended calibration; protected nodes carry the noise penalty. HARP's population ECE can beat undefended none because train-time ϕ-alignment/HCAG reduces overconfidence on the clean majority; on Chameleon, HARP Acc can slightly exceed undefended Acc for the same reason. Across datasets, the portable quality property is the clean majority and clean-slice ECE (Table II). Equal-mass population ECE can look competitive when weak noise regularizes overconfidence (Table IV), but every response is still perturbed.

TABLE I

Cora GraphSAGE systems comparison (5 seeds). Headline at matched Mass$\approx$325: HARP ECE 0.033 $[0.028, 0.038]$ vs. equal-mass 0.159 $[0.153, 0.164]$; ΔAcc vs. strong LBP +0.125 $[0.113, 0.136]$. HARP LiRA TPR@1% FPR 0.011 (none 0.033). †Clip+noise DP-SGD: Acc-tuned point with vacuous $\varepsilon\approx10^3$ (not GAP). ‡GTD/MaskArmor: train-time defenses only—no release Laplace budget, so Mass/Frac are N/A (— — —). §MemGuard perturbs every response (ExactFrac$=0$); Mass is ℓ_1 perturbation mass.

Defense	Acc	LiRA	ECE↓	Mass	Frac
none	0.877	0.567	0.051	—	—
LBP b=0.3	0.707	0.517	0.142	812	1.00
LBP equal-mass	0.870	0.520	0.159	325	1.00
MemGuard§ [6]	0.879	0.538	0.122	—	1.00
GTD‡ [21]	0.878	0.538	0.091	—	—
MaskArmor‡ [22]	0.877	0.580	0.123	—	—
DP-SGD†	0.784	0.509	—	—	1.00
HARP	0.832	0.533	**0.033**	325	0.40
HARP-mask	0.885	0.639	0.092	0	0.40

TABLE II

Clean-slice ECE under locked HARP (3 seeds; Cora aligns with the 5-seed slice study).

Dataset	$\text{ECE}_{\text{clean}}$	ECE_{prot}	ECE_{pop}	none ECE
Cora	0.020	0.090	0.037	0.053
Citeseer	0.083	0.121	0.094	0.140
Chameleon	0.273	0.186	0.238	0.290
Actor	0.254	0.277	0.262	0.262

TABLE III

ΔAcc (HARP−strong LBP) under locked Frac=0.40.

Dataset	ΔAcc	Mass_{LBP}	$\text{Mass}_{\text{HARP}}$	Frac
Cora	+0.125	812	325	0.40
Citeseer	+0.096	998	399	0.40
Chameleon	+0.061	683	273	0.40
Actor	+0.040	2280	912	0.40
PubMed	+0.070	5915	2366	0.40
Photo	+0.185	2295	946	0.40

TABLE IV

Matched-Mass: equal-mass LBP (b=0.12) vs. HARP (3 seeds; Cora: 5).

Dataset	HARP Acc	Eq Acc	HARP ECE	Eq ECE	Clean?
Cora	0.832	0.870	0.033	0.159	HARP 60%
Citeseer	0.716	0.740	0.094	0.073	HARP 60%
Chameleon	0.452	0.439	0.238	0.214	HARP 60%
Actor	0.327	0.324	0.262	0.244	HARP 60%

TABLE V
HARP PRIMARY GRID (GRAPHSAGE; CORA 5-SEED, OTHERS 3-SEED). TPR = LIRA TPR@1% FPR.

Dataset	Defense	Acc	LiRA	TPR	Mass	Frac
	none	0.877	0.567	0.033	—	—
Cora	LBP b=0.3	0.707	0.517	0.019	812	1.00
	HARP	0.832	0.533	0.011	325	0.40
	none	0.744	0.591	0.042	—	—
Citeseer	LBP b=0.3	0.620	0.523	0.030	998	1.00
	HARP	0.716	0.548	0.014	399	0.40
	none	0.448	0.618	0.059	—	—
Chameleon	LBP b=0.3	0.391	0.539	0.038	683	1.00
	HARP	0.452	0.555	0.011	273	0.40
	none	0.335	0.610	0.037	—	—
Actor	LBP b=0.3	0.287	0.549	0.032	2280	1.00
	HARP	0.327	0.561	0.019	912	0.40
	none	0.876	0.505	0.012	—	—
PubMed	LBP b=0.3	0.768	0.499	0.010	5915	1.00
	HARP	0.839	0.502	0.012	2366	0.40
	none	0.802	0.500	—	—	—
Photo	LBP b=0.3	0.552	0.499	—	2295	1.00
	HARP	0.737	0.501	—	946	0.40

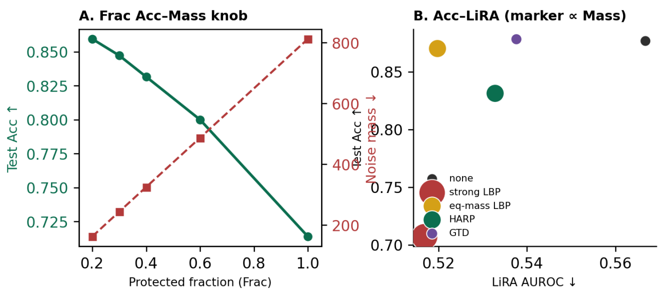

Fig. 3. Frac dial (left) and Acc–LiRA tradeoff with marker size $\propto$ Mass (right). Equal-mass LBP and GTD shown for context.

d) Multi-dataset grid.: Table V repeats the Acc/LiRA pattern on six graphs; PubMed/Photo are near-chance LiRA controls where HARP still preserves Acc vs. strong LBP. Reading down each triple: HARP consistently recovers 4–19 Acc points vs. strong LBP while LiRA moves modestly (often +0.01–+0.02 vs. strong LBP, still below undefended). At the security-relevant low-FPR point, HARP also cuts TPR@1% FPR relative to undefended (Cora $0.033 \to 0.011$; Chameleon $0.059 \to 0.011$; Citeseer $0.042 \to 0.014$). The heterophilic rows (Chameleon, Actor) show the largest undefended LiRA and the clearest case for spending a Frac budget; the near-chance rows (PubMed, Photo) show that HARP's Acc benefit does not require a population LiRA win—operators audit first, then decide whether Frac>0 is warranted.

e) Frac dial and Mass accounting.: Fig. 3 and Table VI sweep Frac on Cora (5 seeds): Acc falls and LiRA improves as Frac rises. Frac is the Acc–Mass dial under fixed σ_{strong}—the systems analog of an empirical privacy spend knob.

f) Exact scores $\neq$ membership safety.: Clean nodes are well calibrated (ECE$_{\text{clean}}$=0.018), but they are not automatically hard to attack. Under locked HARP, unprotected-slice confidence AUROC is $\approx$0.64 vs. $\approx$0.51 population-wide (Fig. 4; Table XI). Raise Frac or enforce session budget B if clean-slice canaries drift up. Aerni et al. [37] motivate such canary checks (Table XIV(a)).

TABLE VI
HARP FRAC SWEEP ON CORA (5 SEEDS; σ_{strong}=0.30).

Frac	Acc↑	LiRA↓	Mass↓
0.20	0.859 [0.855, 0.866]	0.537 [0.532, 0.543]	162
0.30	0.847 [0.843, 0.853]	0.538 [0.533, 0.543]	244
0.40	0.832 [0.826, 0.839]	0.533 [0.525, 0.541]	325
0.60	0.800 [0.793, 0.809]	0.519 [0.512, 0.526]	487
1.00	0.714 [0.699, 0.727]	0.511 [0.503, 0.519]	812

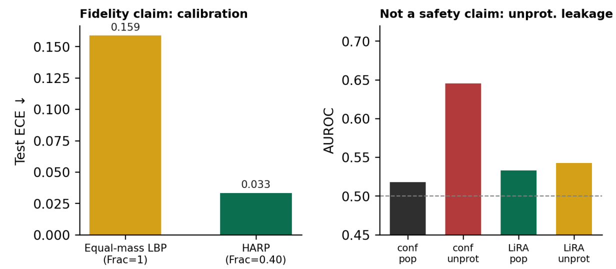

Fig. 4. Veracity vs. safety on Cora. Left: ECE at matched Mass. Right: unprotected-slice confidence AUROC remains 0.645.

TABLE VII
MULTI-QUERY LIRA ON CORA GRAPHSAGE (3 SEEDS; n_{shadows}=4).

Defense	K=1 LiRA	K=5 LiRA	K=20 LiRA
none	0.568	0.568	0.568
HARP	0.534	0.546	0.550
eq-mass LBP	0.520	0.520	0.520

TABLE VIII
SESSION BUDGET ON HARP CORA AT K=20 (3 SEEDS). B=1 FREEZES ACC/ECE NEAR THE ONE-SHOT RELEASE; B=20 MATCHES UNCAPPED AVERAGING.

Policy	B	Acc	LiRA
Budget reuse	1	0.836	0.544
Budget reuse	5	0.847	0.544
Budget reuse	20	0.886	0.548
Uncapped avg	—	0.886	0.548

g) Velocity: multi-query and session budgets.: Averaging K independent releases denoises Laplace noise and moves the API toward the undefended baseline (Table VII). On Cora, HARP LiRA rises from 0.534 at K=1 to 0.550 at K=20, while HARP accuracy rises from 0.836 to 0.886; equal-mass LiRA stays at 0.520 because every node is always noised. A session policy of at most B fresh draws per node—reusing the last draw thereafter—caps this drift (Table VIII). Sweeping $B \in \{1, 5, 20\}$ at K=20 shows B=1 freezes the one-shot defense (recommended default when velocity is adversarial); B=5 is a mild relaxation for benign re-queries. HARP without a session budget is a one-shot defense; Frac and B are complementary velocity controls.

h) Component ablations (consolidated).: Table IX answers "what matters?" in one place. Release-only HARP recovers most Acc vs. strong LBP—selectivity carries the utility win; $k \in \{0, 1, 2\}$ barely moves population LiRA at

TABLE IX
CONSOLIDATED ABLATIONS AT MATCHED FRAC≈0.40 (3 SEEDS). TOP: CORA VARIANTS. BOTTOM: CONSTRUCTOR ACC/LIRA (LTE / RANDOM / AUDIT).

Cora variant	Acc	LiRA	Mass	Train s
LBP b=0.3	0.716	0.517	812	0.8
HARP (locked, k=1)	0.836	0.534	325	2.5
HARP k=0 / k=2	0.834 / 0.828	0.524 / 0.529	325 / 334	2.4
HARP release-only	0.830	0.539	325	0.9
HARP random seeds	0.826	0.526	325	2.0
HARP (GAT) [29]	0.816	0.525	325	1.0
Constructor	Acc / LiRA at Frac=0.40			
	LTE	**Random**	**Audit**	
Cora	0.836 / 0.534	0.818 / 0.522	0.830 / 0.528	
Citeseer	0.716 / 0.548	0.700 / 0.540	0.715 / 0.540	
Chameleon	0.452 / 0.555	0.467 / 0.556	0.463 / 0.561	
Actor	0.327 / 0.562	0.319 / 0.565	0.326 / 0.562	

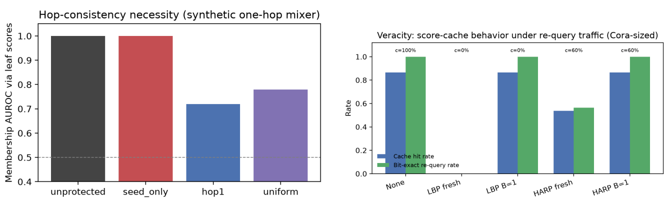

Fig. 5. Left: hop necessity (Prop. 1)—seed-only leaves leaf AUROC at 1.0; hop-1 lowers it to ≈0.72. Right: score-cache hits under re-query traffic (Table X).

TABLE X
SCORE-CACHE SIMULATION (CORA-SIZED; EXACTFRAC c IS BY DESIGN).

Policy	c	Hit rate	Bit-exact re-query
None	1.00	0.865	1.000
LBP fresh	0.00	0.000	0.000
LBP B=1	0.00	0.865	1.000
HARP fresh	0.60	0.537	0.563
HARP B=1	0.60	0.865	1.000

matched Frac (hop geometry, not a population lever). Constructors: LTE beats random seeds on LiRA for heterophilic Chameleon/Actor; on Cora, random is slightly better on LiRA while LTE gains Acc; audit-guided ranking sits between and is preferred only when the LiRA shadow ensemble is already paid for. GAT matches the Acc–Mass pattern (Cora Acc $0.692 \to 0.816$).

i) Why ExactFrac matters (cache simulation).: A Cora-sized synthetic API (800 clients × 25 re-queries; exact-match score cache) makes veracity concrete (Fig. 5, right; Table X): fresh LBP hits 0%; HARP hits ≈54% from ExactFrac alone; B=1 freezes both, but only HARP keeps c=0.60 for consumers needing bit-identical softmax vectors.

j) Qualitative failure case (Chameleon).: Population metrics hide mechanism. Of 375 protected train nodes (seed 42), only 13 (≈3.5%) failed to shrink their train−test true-class confidence gap under HARP—overall mean gap falls $0.46 \to 0.11$. Manual inspection of those 13 failures shows a recurring pattern: Lap(σ=0.3) plus renormalization can *increase* true-class confidence on some draws (e.g., node 177: $0.78 \to 1.00$; node 381: $0.34 \to 0.81$), concentrating mass on the true class by chance. Peaky clean posteriors (conf>0.9) are also hard to flatten. The takeaway for operators: HARP softens the mid-confidence mass that drives population LiRA, but a small overconfident/unlucky tail remains. Raise Frac or σ on that slice if canaries demand it; do not expect uniform-LBP-level erasure at ExactFrac=0.60.

TABLE XI
LTE-AWARE ADAPTIVE QUERIES ON CORA HARP (3 SEEDS).

Query set	Conf@0.4	Conf@0.6	LiRA@0.4
Population	0.513	0.471	0.534
Unprotected (clean)	0.638	0.632	0.539
Boundary (clean)	0.626	0.634	0.535
Top-decile LTE	0.651	0.505	0.532

k) Adaptive LTE-aware queries.: Table XI: querying unprotected nodes, boundary cleans, or top-decile LTE seeds raises conf AUROC $0.513 \to 0.638$–0.651 under locked HARP, while subset LiRA stays near population (≈0.53–0.54). Population LiRA understates conf-MIA on the clean majority—log those canaries and raise Frac or B if they drift. Raising Frac to 0.60 drops the top-decile LTE canary ($0.651 \to 0.505$), but residual clean-slice conf AUROC stays elevated (≈0.63); pair Frac escalation with B=1.

l) Inductive edge exclusion.: Dropping test-incident edges (Cora; 3 seeds) raises undefended LiRA to 0.631 (TPR@1% FPR 0.036). HARP recovers Acc $0.486 \to 0.620$ vs. strong LBP at Frac=0.40 (LiRA 0.569; ECE 0.037). At Frac=0.60, Acc falls to 0.549 while LiRA improves to 0.559—inductive audits may warrant a higher Frac on the same dial.

D. When to spend: volume × structure

Table XII(a): undefended LiRA concentrates in small/mid heterophilic cells; Photo/PubMed/ogbn-arxiv sit near chance under matched shadows—audit first, then set Frac. Table XII(b) reports the honest shadow-count story on Cora for none / LBP / HARP. Absolute LiRA rises with n_{shadows} for every defense [4]; Acc is invariant. HARP does *not* beat strong LBP on LiRA at large shadow counts (Cora n_{sh}=64: LBP 0.561 vs. HARP 0.677)—expected when ExactFrac= 0 is allowed. What HARP keeps is (i) strictly lower LiRA than undefended and (ii) Acc recovery vs. strong LBP (ΔAcc +0.12, invariant in n_{sh}). TPR@1% FPR follows the same pattern.

E. Canaries and large-graph volume

Table XIV(a): HARP suppresses planted/top-decile confidence cues relative to none/LBP; population LiRA remains primary. Table XIV(b): on ogbn-arxiv (∼169k; NeighborLoader; gate off, warmup 3) [28], strong LBP drops Acc to 0.179 $[0.175, 0.184]$. HARP recovers 0.599 $[0.597, 0.601]$ (paired ΔAcc +0.420 $[0.414, 0.425]$; 3 seeds) at Frac=0.40. Defense-aware LiRA stays near chance at both n_{sh}=2 and n_{sh}=4—a volume Acc-recovery result under an audit-null

TABLE XII
AUDIT CONTROLS (GRAPHSAGE; 3 SEEDS). ⋆HETEROPHILIC. (B) COMPARATIVE SHADOW SCALING (NOT HARP-ONLY).

(a) Undefended LiRA (n_{sh}=4)

Dataset	n	Acc	LiRA
Chameleon⋆	2277	0.449	0.623
Cora	2708	0.879	0.568
Actor⋆	7600	0.335	0.510
Photo	7650	0.825	0.500
PubMed	19717	0.876	0.505
ogbn-arxiv	169343	0.666	0.503

(b) Cora shadow scaling (LiRA / TPR@1%)

n_{sh}	none	LBP	HARP
4	0.568 / 0.033	0.517 / 0.022	0.534 / 0.014
16	0.808 / 0.201	0.555 / 0.008	0.670 / 0.037
64	0.847 / 0.265	0.561 / 0.008	0.677 / 0.075

Acc invariant: none 0.879, LBP 0.716, HARP 0.836. Chameleon mirrors the pattern (at n_{sh}=64: none 0.965, LBP 0.685, HARP 0.811; ΔAcc vs. LBP +0.061).

TABLE XIII
OGBN-PRODUCTS: SYSTEMS PROBE (FULL GRAPH) AND BFS-INDUCED 15K-NODE LIRA AUDIT (829K EDGES; 3 SEEDS; n_{sh}=2).

Setting	n	Acc	LiRA	Mass / RSS
Load graph	2.45M	—	—	RSS ≈6.2 GB
Train none (subsample Acc)	2.45M	0.695	—	RSS ≈7.4 GB
Strong LBP budget	2.45M	—	—	Mass $\approx 7.3\times10^5$
HARP budget (Frac=0.40)	2.45M	—	—	Mass $\approx 2.9\times10^5$
BFS-sub none	15k	0.874	0.506	—
BFS-sub LBP b=0.3	15k	0.343	0.500	—
BFS-sub HARP	15k	0.686	0.503	—

TABLE XIV
CANARIES (LEFT) AND OGBN-ARXIV VOLUME (RIGHT; 3 SEEDS). LIRA SHOWN FOR n_{sh}=2 (PRIMARY) AND n_{sh}=4 (STRESS).

(a) Cora canaries

Def.	LiRA	Top	Plt.	ECE
none	0.568	0.551	0.527	0.053
LBP	0.517	0.548	0.543	0.146
HARP	0.534	0.266	0.423	0.037

(b) ogbn-arxiv

Def.	Acc	LiRA_2	LiRA_4	Mass
none	0.666	0.502	0.503	—
LBP	0.179	0.498	0.501	50803
HARP	0.599	0.514	0.505	20322

regime, not a claim that ogbn needs Frac>0. On ogbn-products (∼2.4M nodes) [28], Table XIII reports a systems probe on a 16 GB host plus a defense-aware LiRA audit on a BFS-induced 15k-node subgraph (829k edges; n_{sh}=2). HARP recovers Acc $0.343 \to 0.686$ vs. strong LBP while LiRA stays near chance. Full-graph products LiRA exceeds host memory; the BFS subsample closes that gap without inventing a full-graph privacy claim. The Big Data boundary is clear: release accounting and loader serving scale; full-graph shadow audits need more RAM or distributed training.

If audit LiRA≈0.5, prefer Frac=0; if LiRA$\gg$ 0.5 or veracity requires ExactFrac$\geq c$, set Frac on val Acc (Frac=0.2 maximizes Acc on Cora/Chameleon; Frac=0.4 is the locked compliance default with 60% exact responses).

a) Systems microbenchmarks.: Cora LTE+hop setup ≈41 ms; HARP release ≈0.51 ms vs. LBP ≈0.71 ms; products serving $\approx 3.3\times10^3$ QPS at ≈7.4 GB RSS.

b) Operator workflow.: Audit with defense-aware LiRA; log Mass/Frac/ECE/unprot-conf. If LiRA≈0.5, set Frac=0; else select Frac on validation Acc (Table VI), enforce session budget B=1 by default (Table VIII), and use GAP when a legal ε is required [13]. On Cora/Chameleon, val Acc selection recovers the Acc-maximizing Frac in the grid (Frac=0.2 matches the test oracle); LiRA or ExactFrac constraints instead pick a higher Frac on the same Pareto frontier. Re-audit after model or graph updates.

F. Protocol limitations

Equal-mass and MemGuard can beat HARP on Acc–LiRA when ExactFrac$=0$ is allowed; selective masking raises LiRA; multi-query needs B=1 for a one-shot guarantee; adaptive conf AUROC remains elevated on residual clean nodes even at Frac=0.60 (Table XI); HARP is not DP; ogbn-products full LiRA needs more RAM/distributed shadows.

VIII. DISCUSSION

A. What HARP is for

HARP is for teams that serve GNN scores as a data product and need three things at once: usable accuracy, a clean majority of reproducible probabilities (ExactFrac$\geq c$), and an auditable noise budget. It is *not* a drop-in replacement for formal differential privacy: when a legal ε is required, use training-time DP such as GAP [13], [16]. Membership audits (LiRA) tell the operator whether Frac should be zero or positive; Acc and ECE tell them what the chosen Frac costs. When LiRA≈0.5, the systems recommendation is Frac=0. When policy still mandates strong σ *or* ExactFrac$\geq c>0$, selective release is the feasible class; uniform LBP is not (Remark 1).

B. Impact for Big Data graph APIs

Hop-consistent release sets are the analogue of topology-aware row budgets; Mass/Frac give spend telemetry. ExactFrac enables score-cache hits that uniform noise cannot (Table X); heterophilic graphs recover Acc vs. b=0.3; audit-null large graphs argue against blanket noise.

C. Threats to validity

Cora fairness/Frac: five seeds with paired CIs; other cells: three seeds. Mass/Frac are spend metrics, not (ε, δ); ExactFrac/ECE measure fidelity, not per-node hardness. Products subsample LiRA and ogbn-arxiv n_{sh}=4 extend scale audits without claiming full-graph products privacy.

D. Relation to formal differential privacy

Prop. 3: finite local ε on the protected minority, ε=∞ on the clean majority—hence infinite global ε whenever ExactFrac>0. That incompatibility is the constrained-release tradeoff. Use GAP for regulatory DP; use HARP when ExactFrac$\geq c$ matters.

E. What we do not claim

HARP is not a stronger membership defense than uniform noise; does not dominate equal-mass/MemGuard when ExactFrac$=0$; hop k is not a monotone LiRA lever; multi-query without B erodes the one-shot defense; clean-slice conf-MIA stays elevated unless Frac or B rises (Table XI).

F. Future work

Adaptive Frac from streaming audits, co-optimizing formal ε with Mass [31], [32], federated hop-respecting APIs, and distributed LiRA at products/papers100M scale.

IX. Conclusion

We proposed HARP, a selective release framework for GNN prediction APIs under constrained score release: maximize accuracy subject to a membership audit and a minimum exact-response fraction. HARP is not a stronger membership defense than uniform noise; it is the feasible mechanism under ExactFrac$\geq$ $c>0$, with hop-consistent allocation (Prop. 1), selective local-ε accounting (Prop. 3), and Acc/calibration recovery versus strong LBP—including a quantified cache-hit benefit from the clean majority. On six benchmarks plus OGB scale-ups with defense-aware LiRA, membership audits guide *when* to spend budget; Frac and B are the operator dials. Use training-time DP when a legal global ε is required.

a) Reproducibility.: Code (`defenses/harp.py`, `defenses/memguard.py`, `run_harp_eval.py`), locked configs, and frozen CSVs accompany the submission artifact at https://github.com/krishharma/privacy-gnn. Runs used Darwin arm64 CPU; LiRA n_{shadows}=4 primary (16/64 comparative sweeps on Cora/Chameleon).

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